Thesis Defence

Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

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Thesis Structure

1 Introduction
- 1.1 Motivation & Background
- 1.2 The Research Gap
- 1.3 Research Questions & Hypotheses
- 1.4 Contributions to the Literature
- 1.5 Thesis Outline

2 Literature Review
- 2.1 Theories of Commodity Futures Returns
- 2.2 Aggregate Commodity-Specific Factors
- 2.3 Individual Commodity Fundamentals
- 2.4 Macroeconomic Predictors
- 2.5 ML Methods in Return Forecasting
- 2.6 Volatility Regimes and Predictability

3 Methodology
- 3.1 Baseline Framework (Replication)
- 3.2 Extension I: Penalized Regression
- 3.3 Extension II: Individual Fundamental Predictors
- 3.4 OOS Evaluation Criteria
- 3.5 Portfolio Back-Testing Framework

Structure

- There are several key implications, that can be drawn from the thesis.

Model	RMSE	R2
OLS	0.421	0.120
LASSO	0.387	0.180
Elastic Net	0.372	0.210

Table 1: *Model comparison*

Thank you for your attention!